

Modeling Reddit-Discussions on Disability

Using BERTopic to Identify Themes in the r/Disability Community

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Abstract

This study analyzes discussions on the r/Disability subreddit from July to October 2025 to identify prevailing topics and examine temporal trends. Using BERTopic, a transformer-based topic modeling approach, 3,200 posts were clustered into 12 coherent themes, with outliers minimized through post-processing. Topic engagement metrics, including post counts, scores and comments, were analyzed across months to assess both long-term and event-driven discussion patterns. Results indicate a mix of persistent themes, such as mobility and general health and short-term spikes associated with seasonal events or policy debates. The highest engagement of the community was measured for posts related to personal experiences or high-impact news. These findings highlight how user interests within the disability community evolve over time and suggest that combining topic frequency with engagement metrics provides a richer understanding of public attention shifts. Along with the analysis, a tool was created with the help of Gradio, such that anyone can run the same analysis at any time.

1 Introduction

Online discussion platforms such as Reddit provide a rich source of user-generated content that reflects personal experiences, opinions, and challenges. In particular, communities focused on disability topics contain diverse narratives about accessibility, health, mobility, and social inclusion. To better understand the themes that emerge within this discourse, we applied topic modeling to a dataset of posts collected from the r/Disability subreddit. Using the BERTopic [4] framework, which combines transformer-based document embeddings with clustering algorithms, we aimed to identify coherent topics and explore how users engage with them. This analysis allows us to uncover recurring concerns and perspectives shared by members of the disability community, offering insights into the most discussed issues online.

2 Methods

2.1 Data Collection

The dataset used in this project was collected from the r/disability subreddit on Reddit. This community focuses on discussions on the topic disability. There, one can find posts to various themes such as accessibility, living with disabilities, assistive tools like wheelchairs, or social experiences of disabled individuals. Reddit provides a large amount of user-written data that will help us get a perspective on the most important topics. Posts were retrieved using the PRAW (Python Reddit API Wrapper) [6] interface. A downside to using this API is, that we were limited to only retrieve 1000 posts at a time. The same data collection tasks were tested with PushShift API [2], but because this tool is now outdated, we chose PRAW. The scraper was configured to collect posts from the r/disability subreddit using different filters. "Top-mode" prioritizes posts with the highest engagement (measured by their upvote/downvote ratio) in a given time frame, "hot-mode" uses a similar approach but weights the newer votes higher than older ones, effectively returning the most trending posts. "New-mode" simply retrieves the newest posts. For each submission, the following metadata fields were stored for later use: Post ID, Title, Body text (selftext), Score (upvotes) and Comment count.

2.2 Preprocessing

Before topic modeling, extensive text preprocessing was applied to ensure that the input to the language model was clean and meaningful for the model. To get valid results from our model we used the following preprocessing methods:

- All posts were transformed to lowercase
- All posts were stripped of punctuation, URLs and special characters (numbers were left in since they sometimes provide meaningful data)
- For shorter texts, the title was added to the selftext
- Posts that were still shorter than 30 characters were discarded
- Stopword removal was performed using a combined stopwords list from Spacy, SKLEARN [5] and a custom list containing common Reddit-specific terms (e.g., "i'm", "like", "people")
- Lemmatization was applied to reduce inflected words to their base form, improving the overall effectiveness.

This preprocessing pipeline improved clustering stability by reducing noise and vocabulary sparsity.

2.3 Topic Modeling

The core of this analysis relied on BERTopic [4], a transformer-based topic modeling approach that combines contextual embeddings with density-based clustering. Unlike traditional models, BERTopic leverages c-TF-IDF to create dense clusters allowing for easily interpretable topics whilst keeping important words in the topic descriptions. With sentence embeddings (from transformer models such as all-MiniLM-L6-v2) we can capture semantic similarity between sentences and paragraphs. Each post was first converted into an embedding vector, which was then reduced in dimensionality using UMAP to make clustering feasible in low-dimensional space. Clustering was then performed using HDBSCAN, which automatically determines the number of clusters and identifies outliers. To get the best possible results, we had to tweak some parameters. The most important for UMAP were "n_neighbors" - determines how local or global the UMAP representation is, "n_components" - the dimensionality of the reduced embedding space and "metric" - the distance metric used in UMAP. For HDBSCAN the most important parameters were "min_cluster_size" - the minimum number of posts required to form a cluster, "min_samples" - controls how conservative a cluster is and which are labeled as outliers, "metric" - the distance metric used in HDBSCAN. Lastly, for BERTopic we have "top_n_words" - how many keywords BERTopic shows for each topic, "min_topic_size" - minimum number of documents per topic. To achieve better overall results, we also conducted outlier reduction after the modeling, which aims to shift posts that were labeled as outlier into close clusters according to a metric and a threshold. The different metrics are "probabilities", which reassigns posts based on their topic probability vector. The "c-tf-idf"-metric checks, which topic vocabulary best matches the outlier document. Lastly, the "embedding"-metric compares the embedding vector of each outlier to the average vector of the topic and reassigns them. The resulting topics were ranked by frequency, and their representative keywords were extracted using class-based TF-IDF weighting. Each topic was manually reviewed afterwards to verify interpretability.

2.4 Automatic and Manual Labeling

Once the BERTopic model was trained, the resulting clusters were analyzed and labeled using an automatic or a manual approach. For the automatic labeling, an LLM-based summarization method was applied to the top keywords and two representative documents of each topic. Different LLMs were used for this task, namely flan-t5 [3], alpaca [7] and mistral. Out of these three, alpaca performed the best, so we chose that result for the automatic approach. Each model was prompted to generate a short, descriptive label summarizing the theme. For example, the keywords ["pain", "chronic", "fatigue", "illness", "medication"] were labeled as "Chronic Illness and Pain Management." However, since automatic labels can sometimes be inconsistent or overly general, a manual correction interface was developed that can be used from the terminal (and also Gradio, see Section 3). When using the manual labeling approach, the user was shown the top words of the topic, two or three example documents and the labels that were created by the LLMs. After analysing the data manually, the user is able to input a custom label for the topic. The final curated label set is later used for export and visualization. This setup ensures, that the final topic names remain human-readable and semantically appropriate.

2.5 Analysis

The last step in our pipeline is the analysis part, where we aim to summarize and interpret the model results. For this, an engagement table was created, containing the most important data like the topics and their labels, score mean and median per topic, number of comments mean and median per topic, post count, a list of keywords and representative documents per topic. Additionally, several meaningful visualizations using the built-in tools of BERTopic and additional plotting functions were generated. The "visualize_barchart()" function of BERTopic creates an overview of all found topics and its top keywords. To display the relative size (number of posts) of each cluster, the mean score of each topic and the mean number of comments per topic, three additional bar charts were created. Since all these plots show their metric without a clear underlying connection, we wanted to also create a plot that combines all the data into one plot. The cleanest way to show this data will be as a bubble plot. The size of the bubbles resemble the number of documents per topic, and the x- and y-axes represent the mean score and comments per topic. Lastly, some scatterplots were created for each topic individually that show the mean score and comments of every post per topic. The goal of this is so that we can see the distribution of score and comments in the topics.

3 Gradio Interface Tool

To streamline the workflow from data collection to analysis, a comprehensive interactive system was developed using the Gradio [1] framework. This web-based tool provides a modular interface for each stage of the pipeline — from collecting Reddit posts to preprocessing, topic modeling, labeling, and visualization. The system effectively replaces a multi-script workflow with an integrated and user-friendly control panel.

3.1 System Overview

The tool is divided into five main tabs, each representing a distinct stage of the research process. These parts are approximately as described in Section 2 but to quickly summarize, we will explain the different tabs and their functions:

1. **Data Collection:** Uses the PRAW API to fetch posts from the r/disability subreddit. The user can choose between Reddit’s “top,” “new,” or “hot” modes and specify the number of posts. The retrieved posts (title, body, score, and comment count) are stored locally for reproducibility.
2. **Preprocessing:** Provides options for text cleaning, lemmatization, and stopword removal using both Spacy and scikit-learn. Parameters such as minimum text length can be adjusted dynamically to refine the dataset.
3. **Topic Modeling:** Integrates the BERTopic model with configurable parameters for UMAP, HDBSCAN, and topic generation. This allows real-time experimentation with clustering behavior and dimensionality reduction. The system also supports multiple outlier reduction strategies (probability-based, c-TF-IDF, and embedding-based), providing flexibility in refining the topic structure.
4. **Labeling:** Offers both manual and automatic labeling options. In the manual mode, the user can view representative posts and keywords for each topic, and assign concise human-readable labels. The automatic mode uses the flan-alpaca-large model to generate label suggestions based on topic keywords and example posts.
5. **Analysis:** Summarizes and visualizes topic-level engagement data. The system produces plots showing post distribution per topic, average scores and comments, and a bubble plot that compares engagement metrics across topics. Users can preview and export the aggregated dataset as a CSV file for further exploration.

3.2 Design and Usability

The Gradio interface was chosen for its simplicity and flexibility. Each operation runs asynchronously, allowing users to monitor progress through real-time status updates and logs. Visualization outputs, such as topic barcharts or engagement plots, are rendered directly in the interface using integrated Matplotlib and Seaborn components. This modular design ensures that the tool can be used not only for this project but also generalized to other subreddit analyses. The clear separation between tabs allows users to rerun individual steps independently — for instance, updating preprocessing parameters without re-fetching data.

3.3 Benefits

By combining all key components into a single environment, the Gradio system achieves:

- A transparent workflow from data collection to insight generation
- Faster experimentation through parameter sliders and dropdown controls
- Real-time feedback and visualization at every step

Overall, the system acts as both a research framework and an interactive visualization platform, enabling efficient exploration of Reddit discussions around disability-related topics.

4 Results

4.1 Summer Snapshot

This section summarizes the main outcomes of the Reddit discussion analysis conducted over the summer period of 2025. The “snapshot” represents the state of the project after the data pipeline, topic modeling, and labeling processes were completed on a dataset, that was gathered over the summer months June-September.

4.1.1 Dataset Overview

The final dataset contained approximately 3,200 posts collected from the r/disability subreddit. After cleaning, filtering, and removing very short or irrelevant entries, the dataset provided a diverse sample of discussions ranging from medical experiences to accessibility challenges. Furthermore, after preprocessing, our dataset has 2800 entries, which is well within our set boundary of 2000-3000 posts. After configuring our model parameters for a dataset of this size (the exact values for the parameters can be seen in Appendix A), BERTopic identified 12 coherent topics and the outliers (documents not assigned to any cluster) were reduced to 225 posts using the methods mentioned in Section 2.3.

4.1.2 Topic Overview

Figure 1 shows the topic frequency distribution. The y-axis lists the most prominent topics (sorted by size), while the x-axis represents the actual number of posts associated with each. When looking at the results of the most occurring themes on Reddit, we were quite surprised to see "Living, family, romance" at the top because we predicted that posts on accessibility, wheelchairs and other movement aids appear much more on the website. We were pleasantly surprised that certain temporary topics such as "Pride month (June)" and "America's medicaid/medicare cuts" also occur in the results. We also predicted the topic "mental disabilities" to be bigger.

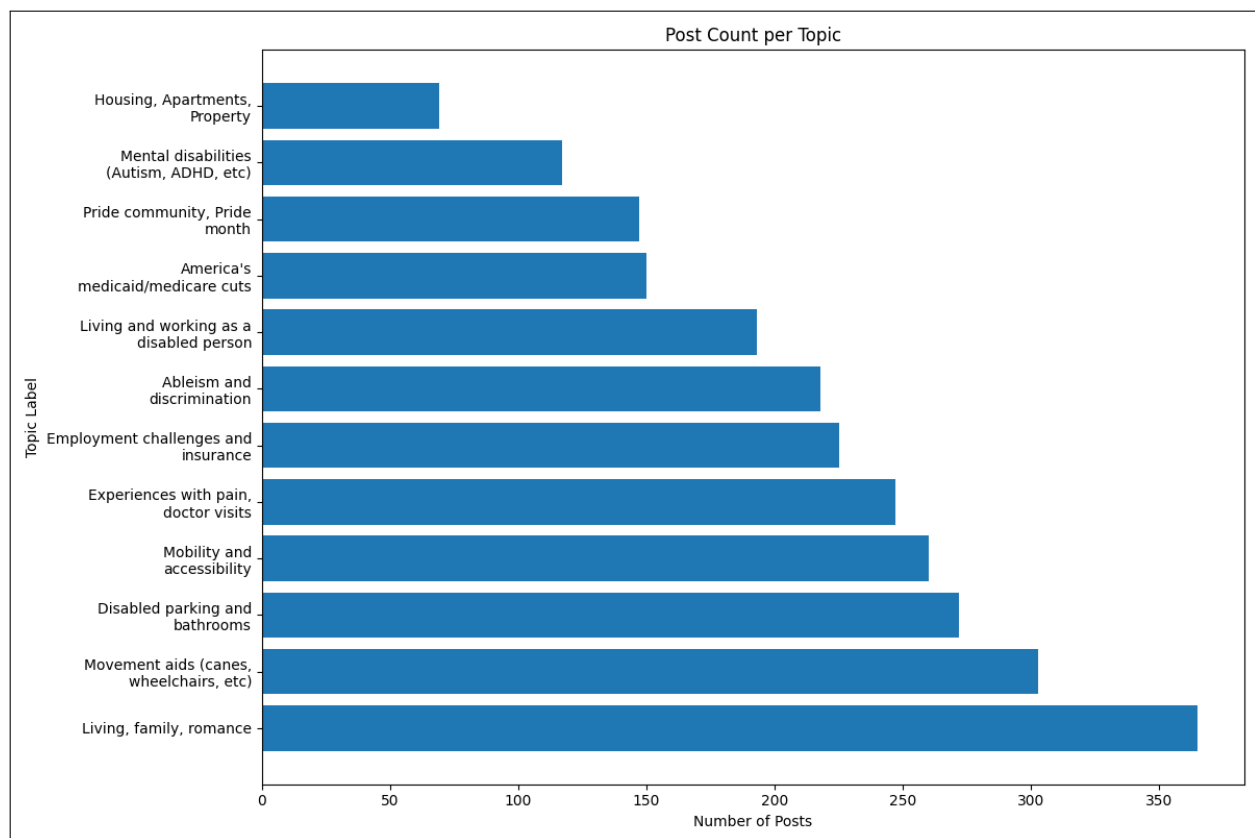


Figure 1: Distribution of topics, sorted by frequency.

4.1.3 Engagement Metrics

Beyond identifying discussion themes, we also analyzed engagement metrics per topic. Figure 2 visualizes the average number of upvotes and comments for each topic, with bubble size corresponding to the number of posts within that topic. Highly engaged discussions were generally associated with personal experience sharing, discrimination and mental health topics, while posts about material things such as movement aids, housing and property tended to attract fewer comments. It was also quite interesting that the most up voted topics were the temporary ones we already discussed above in Section 4.1.2. Also, posts focusing on mobility gained many upvotes. The most engaged topic, even though not the biggest one, was "America's medicaid cuts". In Appendix B, we can also see two bar charts of the most commented on and highest scored topics separately.

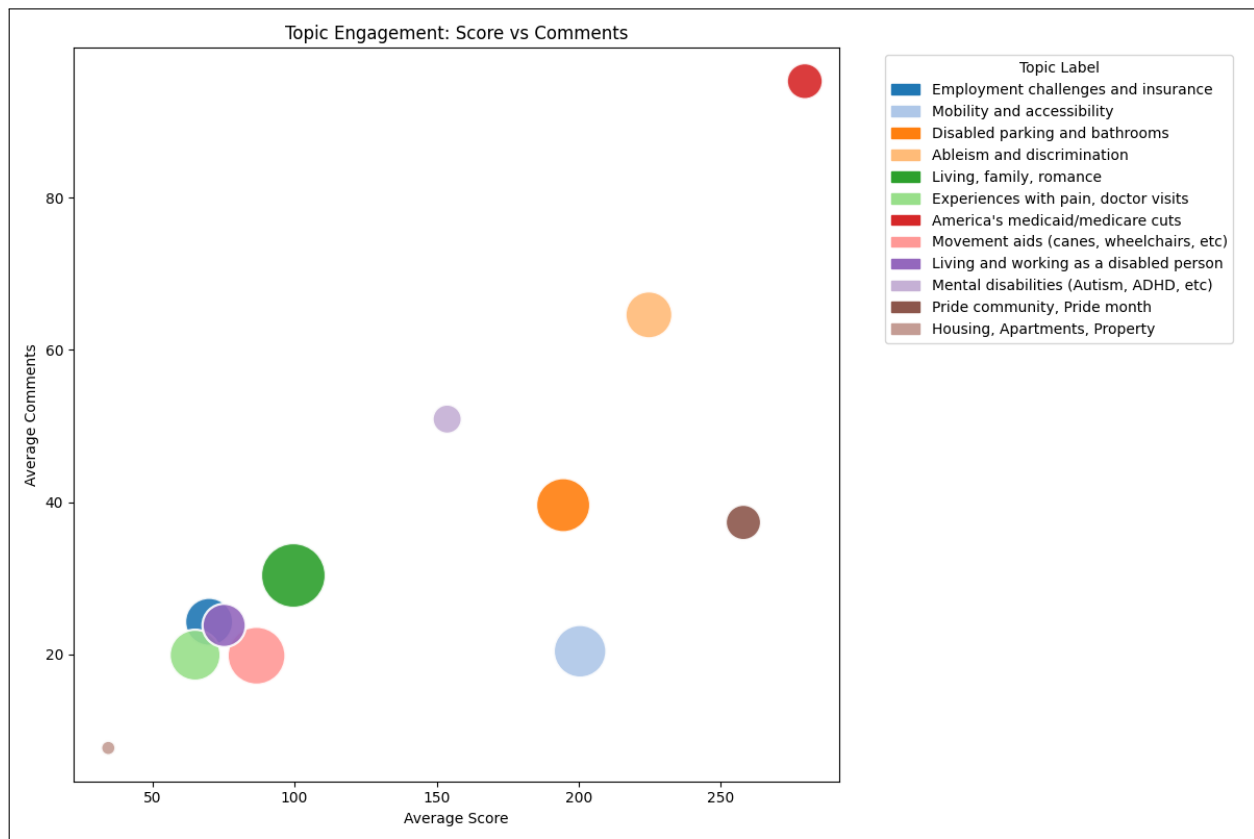


Figure 2: Bubble plot showing mean comment and score distribution per topic.

4.1.4 Representative Topics

Table 1 summarizes the first few entries for the generated csv-file. Each entry includes the final label, mean score, post count, mean number of comments and top keywords. For this table, it has to be noted that the found topics are not ordered by any metric. The actual csv-file, which can be found in the project also features the median score, median number of comments and lastly, some example texts. This overview demonstrates both the variety of themes and the interpretability of the BERTopic results.

Topic Label	Score mean	Post count	Comments mean	Top Keywords
Employment challenges	69.97	225	24.28	work, job, claim, benefits
Mobility, accessibility	200.43	260	20.41	wheelchair, scooter, mobility, chair
Ableism, discrimination	224.66	218	64.59	hate, person, ableism, dont
Living, family, romance	99.60	365	30.36	feel, live, mental, friends

Table 1: Sample of representative topics identified in style of the generated csv-file

4.2 Trend Analysis

To understand how the focus of discussions changed during the summer months, we examined the temporal distribution of topics from July to October 2025. Each topic was sampled on a monthly basis, and the absolute number of posts per topic was visualized in a line chart. This approach allows for tracking the volume of discussions for each topic over time, rather than relative proportions. To get the best results, we used the search tag "top" because the other search tags created too much fluctuation, such that the graph was not interpretable anymore. Furthermore, it should be noted that the Reddit API decreased the size of posts one could get at any point in time from 1000 to 250 total posts, which led to an overall decrease in any metric from October. This approach highlights how user interests shifted over time, independent of changes in total posting volume.

Figure 3 summarizes these trends. Several patterns can be observed:

- **Seasonal or event-driven topics increased sharply at specific points.** Similar to the "Pride" topic we have seen in Section 4.1 already, that was almost exclusively present in June and July, we can

also see some temporary topics here in the line graph. For one, the "Disabled artists" topic spikes in August, which may be caused by an event called "Disability Arts Festival". This event took place in Toronto in August and is therefore likely causing the spike.

- **Stable, long-term themes remained consistent throughout the entire window.** Topics such as "Ableism and discrimination", "Mobility aids, accessibility" and "General health, doctor visits" showed only minor fluctuations over time and are likely to stay this way, since these topics are important every-day issues. These represent ongoing concerns within the disability community that persist independently of external events.
- **Short-lived, high-engagement spikes.** Topics like "America's medicare cuts" were caused by a real time event. This topic is noticeable for its short lived but huge spike. These are topics that are caused by an event but the community can still feel their impact in contrast to the seasonal topics, which usually leave the spotlight earlier.

Overall, these temporal patterns demonstrate that the r/Disability community features a mix of persistent, structural themes but also some short-term topics, that focus on certain events. The combination of monthly relative frequencies and engagement metrics provides a richer understanding of how public attention shifts within disability-related topics. An idea for future work would be to make more data points per month to achieve a smoother line graph.

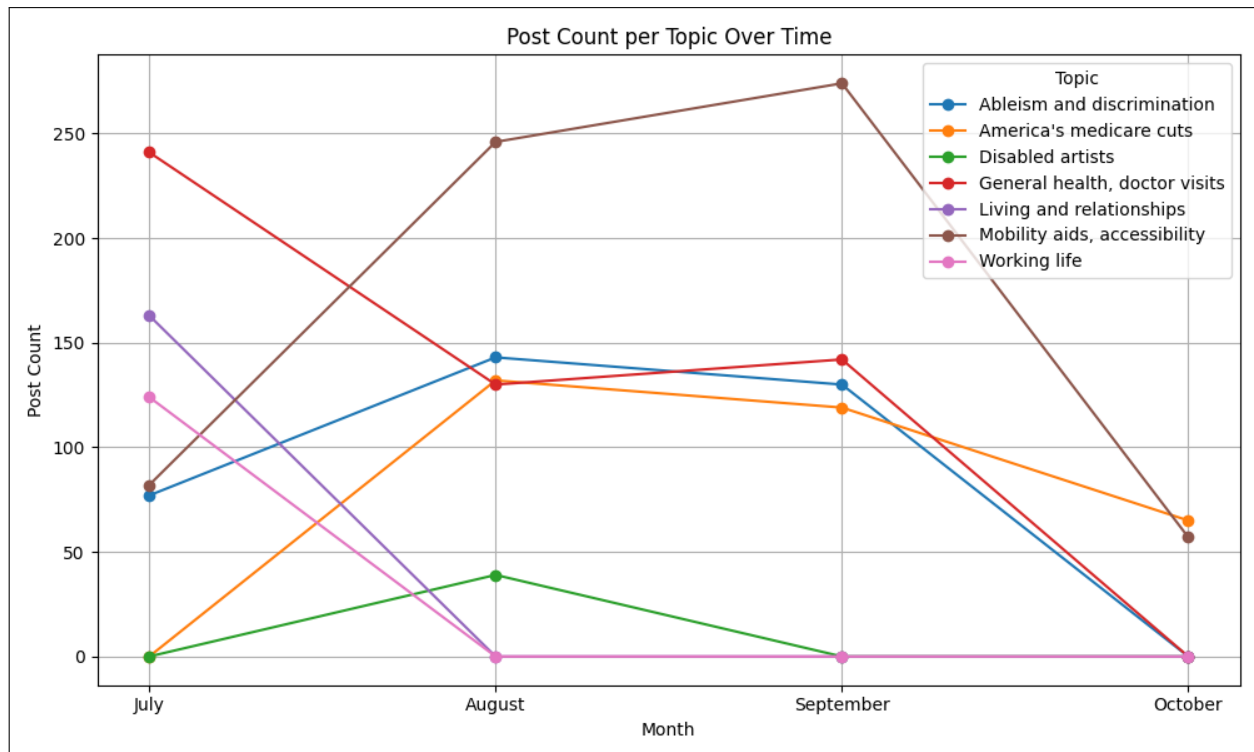


Figure 3: Relative topic frequencies from June to September 2025.

5 Conclusion

The analysis of Reddit posts from July to October 2025 reveals a mix of persistent and event-driven topics within the r/Disability community. Long-term themes such as mobility and general health maintain a consistent level of discussion, while seasonal or event-driven topics, such as Disability Arts Festival or policy debates, cause short-term spikes. Engagement metrics indicate that posts related to personal experiences or high-impact news attract more comments and upvotes, regardless of overall post count. Future work could involve collecting more posts per month and exploring other engagement metrics to better understand user interaction patterns.

A UMAP, HDBSCAN and BERTopic Parameters

Here are the exact parameters used for UMAP and HDBSCAN in Section 4.1.1.

UMAP Param.	Value	HDBSCAN Param.	Value	BERTopic Param.	Value
n_neighbors	21	min_cluster_size	21	top_n_words	20
n_components	5	min_samples	20	min_topic_size	50
metric	cosine	metric	euclidian		

Table 2: Snapshot parameters

B Additional Plots

Additional visualizations of the Snapshot are shown here.

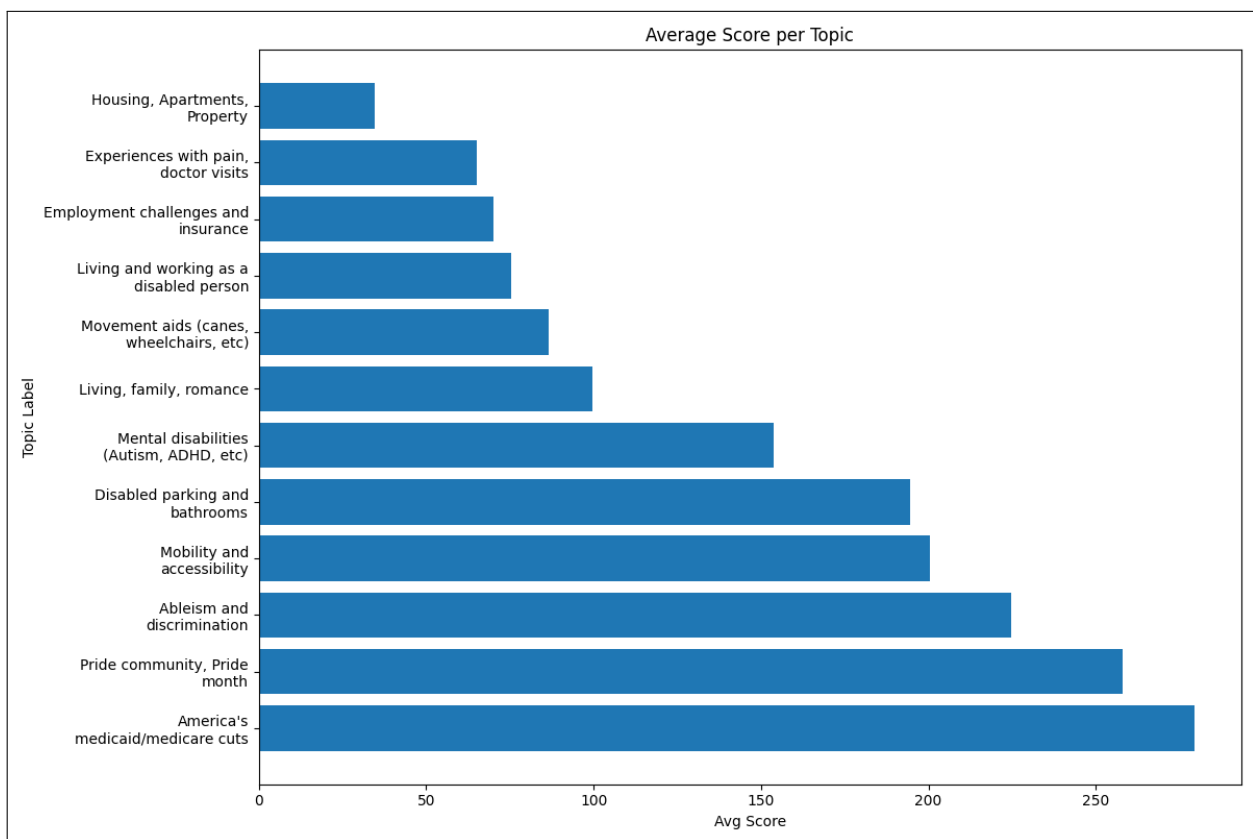


Figure 4: Distribution of topics, sorted by the average score.

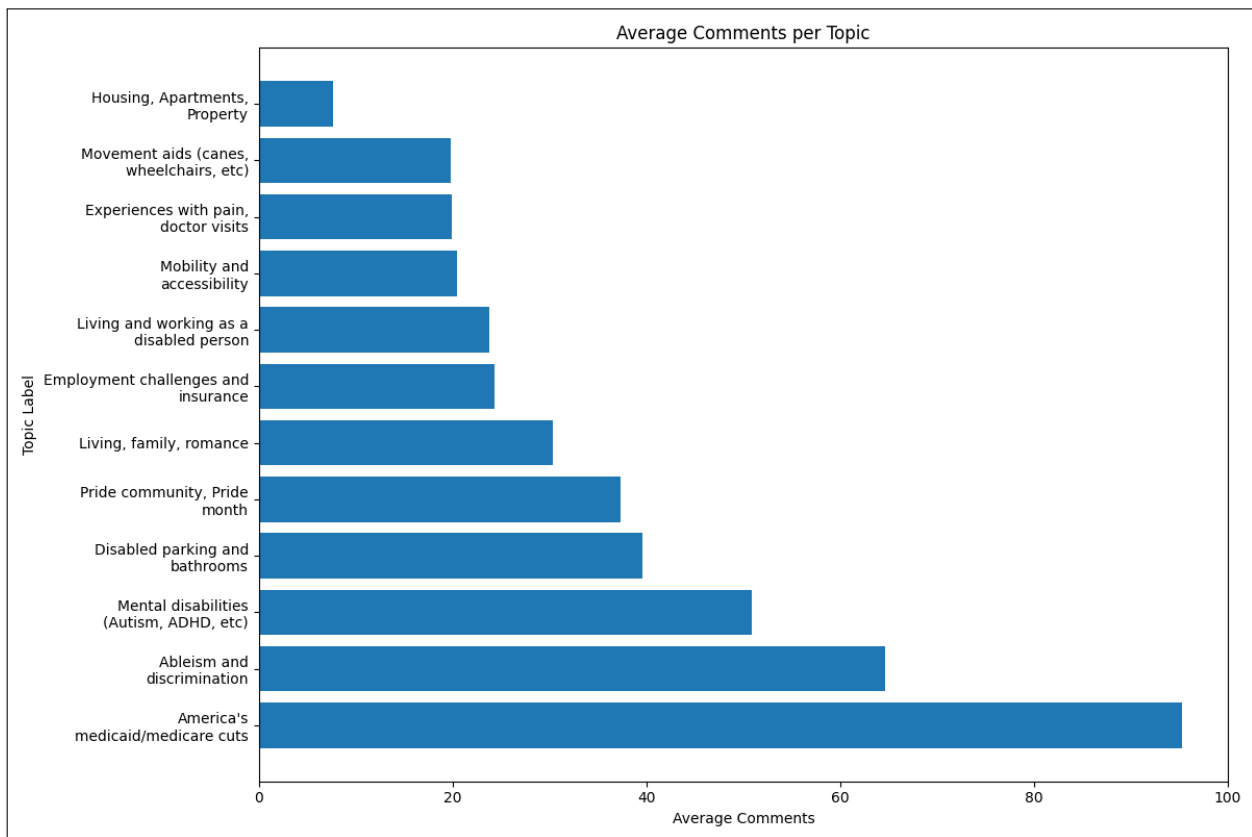


Figure 5: Distribution of topics, sorted by the average number of topics.

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